



B.Tech. Degree VII Semester Special Supplementary Examination April 2025

ME 19-205-0703 MACHINE DESIGN II
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the customers need, formulate the problem and draw the design specifications.
 CO2: Understand component behaviour subjected to loads and identify the failure criteria.
 CO3: Apply the concepts of stress analysis, theories of failure and material science to analyse, design and/or select commonly used machine components.
 CO4: Analyse the stresses and strains induced in a machine element.
 CO5: Design machine components using theories of failure.
 CO6: Get basic knowledge of the preparation of working drawings by including the manufacturing details like tolerance, surface finish etc.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

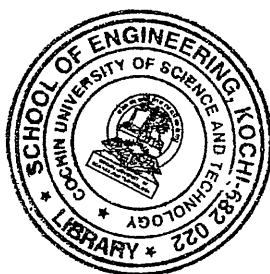
PO – Programme Outcome

(Design data books are permitted for examination)

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	What are the materials used for lining of friction surfaces?	3	L1	2	1
(b)	What is a self-energizing brake? When will a brake become self-locking?	3	L1	2	1
(c)	What is the effect of the following on the dynamic loading of gears? (i) Face width. (ii) Pitch line velocity. (iii) Material elasticity.	3	L2	2	1,2
(d)	Usually, the worm is made of hardened steel and the worm wheel is made of softer material. Why is it so?	3	L2	1	1,2
(e)	Explain the following terms as applied to journal bearings: (i) Bearing characteristic number. (ii) Bearing modulus.	3	L1	2	1
(f)	What are rolling contact bearings? Discuss their advantages over sliding contact bearings.	3	L2	1	1
(g)	Briefly explain the importance of surface finish in the design of machine elements	3	L2	6	1,2
(h)	What are the points to be considered while designing a turned part?	3	L2	6	1,2

(P.T.O.)



PART B

		(4 × 12 = 48)	Marks	BL	CO	PO
II.	A single dry plate clutch is to be designed to transmit 7.5 kW at 900 rpm. Find: (i) Diameter of the shaft. (ii) Mean radius and face width of the friction lining assuming the ratio of the mean radius to the face width as 4. (iii) Outer and inner radii of the clutch plate. (iv) Dimensions of the spring, assuming that the number of springs are 6 and spring index equals 6. The allowable shear stress for the spring wire may be taken as 420 MPa.		12	L3	3	1,2,3
OR						
III.	Two pulleys, one 450 mm diameter and the other 200 mm diameter, on parallel shafts 1.95 m apart are connected by a crossed belt. Find the length of the belt required and the angle of contact between the belt and each pulley. What power can be transmitted by the belt when the larger pulley rotates at 200 rpm, if the maximum permissible tension in the belt is 1 kN and the coefficient of friction between the belt and pulley is 0.25.		12	L3	4	1,2,3
IV.	A 15 kW and 1200 rpm motor drives a compressor at 300 rpm through a pair of spur gears having 20° stub teeth. The centre-to-centre distance between the shafts is 400 mm. The motor pinion is made of forged steel having an allowable static stress as 210 MPa, while the gear is made of cast steel having allowable static stress as 140 MPa. Assuming that the drive operates 8 to 10 hours per day under light shock conditions, find from the standpoint of strength, (i) module. (ii) face width. (iii) number of teeth. (iv) pitch circle diameter of each gear.		12	L3	4	1,2,3
OR						
V.	Design 20° involute worm and gear to transmit 10 kW with worm rotating at 1400 rpm and to obtain a speed reduction of 12:1. The distance between the shafts is 225 mm.		12	L3	3	1,2,3
VI.	Design a main bearing for the compressor to sustain a load of 4.5 kN. The speed is 1500 rpm. Ambient temperature = 25°C. Diameter of the main shaft = 50 mm.		12	L3	3	1,2,3
OR						
VII.	Select a single row deep groove ball bearing for a radial load of 4000 N and an axial load of 5240 N, operating at a speed of 1600 rpm for an average life of 5 years at 10 hours per day. Assume uniform and steady load.		12	L3	5	1,2,3
VIII.	(a) What are the general design recommendations for welded parts? Explain with sketches. (b) Explain with a suitable example the steps in improving the design of a component from the point of view of machining.		8 4	L2 L2	6 6	1,2,3 1
OR						
IX.	(a) Describe the general important considerations for the choice of assembly methods. (b) What are the general design recommendations for rolled sections?		8 4	L2 L2	6 6	1,2,3 1

Bloom's Taxonomy Level's

L1 – 7.50%, L2 – 32.5%, L3 – 60%.